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Frontier by Default Is a Design Flaw

LLMDay NY, 2026

About Me

AI and Creative Technologist with CS, Math, and Cognitive Science background.

Co-Founder of ReFwd AI and creator of Spens - a sandboxed, fully observable agent runtime

Delivered AI Solutions for the following brands:









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Today's Talk

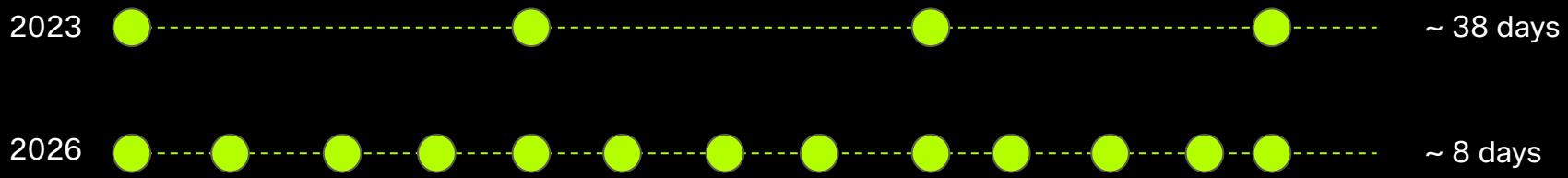
Frontier by default is a design flaw

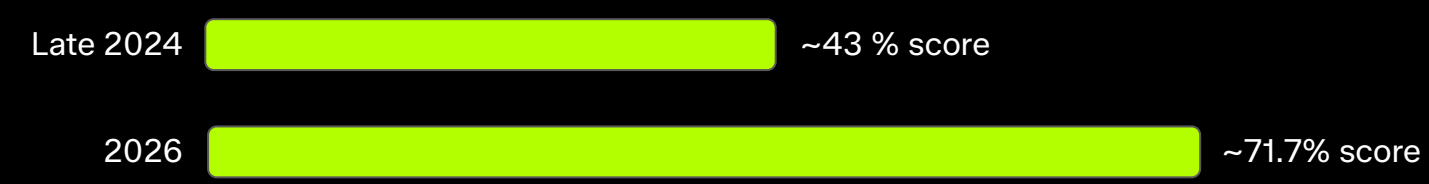
How I learned to love smaller models and hybrid routing

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THE PROBLEM

The Frontier is accelerating





Sources: The Release Clock (2026); Stanford AI Index 2026

Default is good for prototypes but not good for production

Gartner predicts more than 40% of agentic AI projects will be cancelled by the end of 2027 due to escalating costs, unclear business value, or inadequate risk controls

Token economics

Cost per task may outweigh ROI

Privacy and data sovereignty

Data may not be able to leave your infra, region, or device

Connectivity and latency

Some use cases may not have reliable internet

Question: How do we get the best of all the worlds - Speed, Cost, and Accuracy?

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Good News!

SLMs and Open Weights are not frontier. But no longer a toy.

The lag is shortening

In 2024 they lagged > 12 months; now it's closer to 4

Tool calling is reliable

~100% format adherence with constrained decoding

Costs are down

Consumer hardware can provide what the frontier did 12 months ago

Epoch AI (CC BY), 2024 and 2026

Approach

Several Techniques to optimization

Fine Tuning
Task-Specific Models

Local Models
Leveraging local models and hybrid

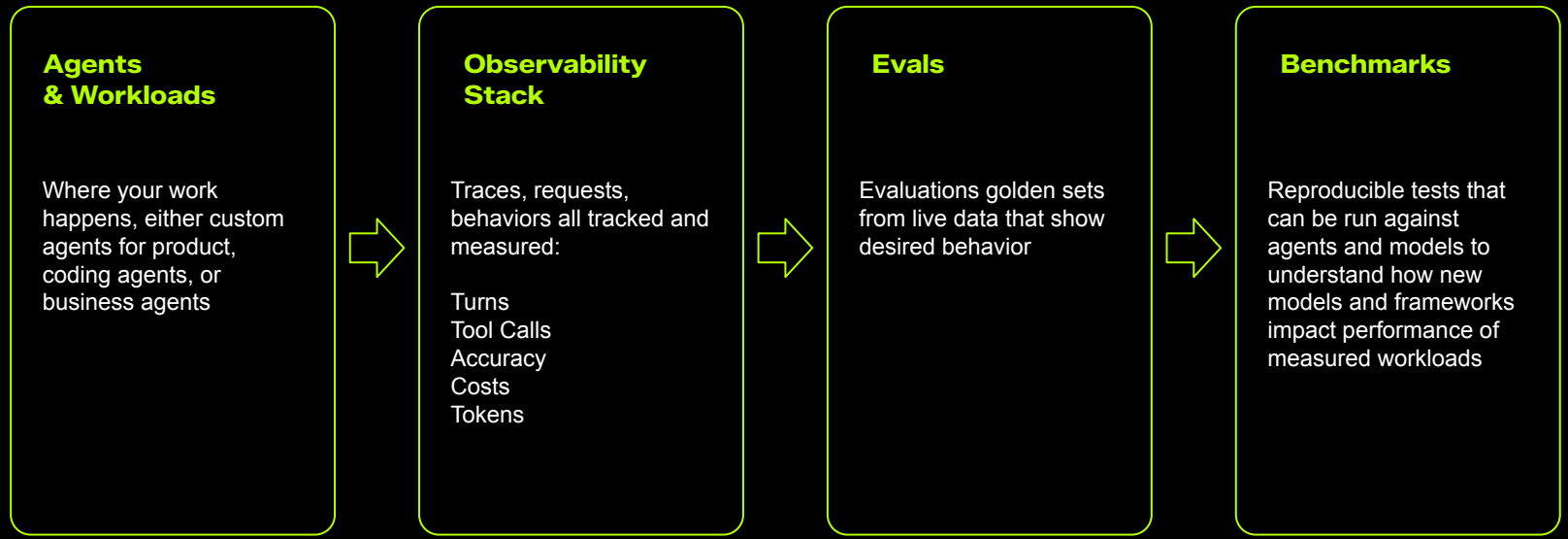
Dynamic Routing
Escalation, decomposition

Just Improve the Frontier
Caching, tool call management, and compaction

Foundational
Evals, Observability, and Benchmarking

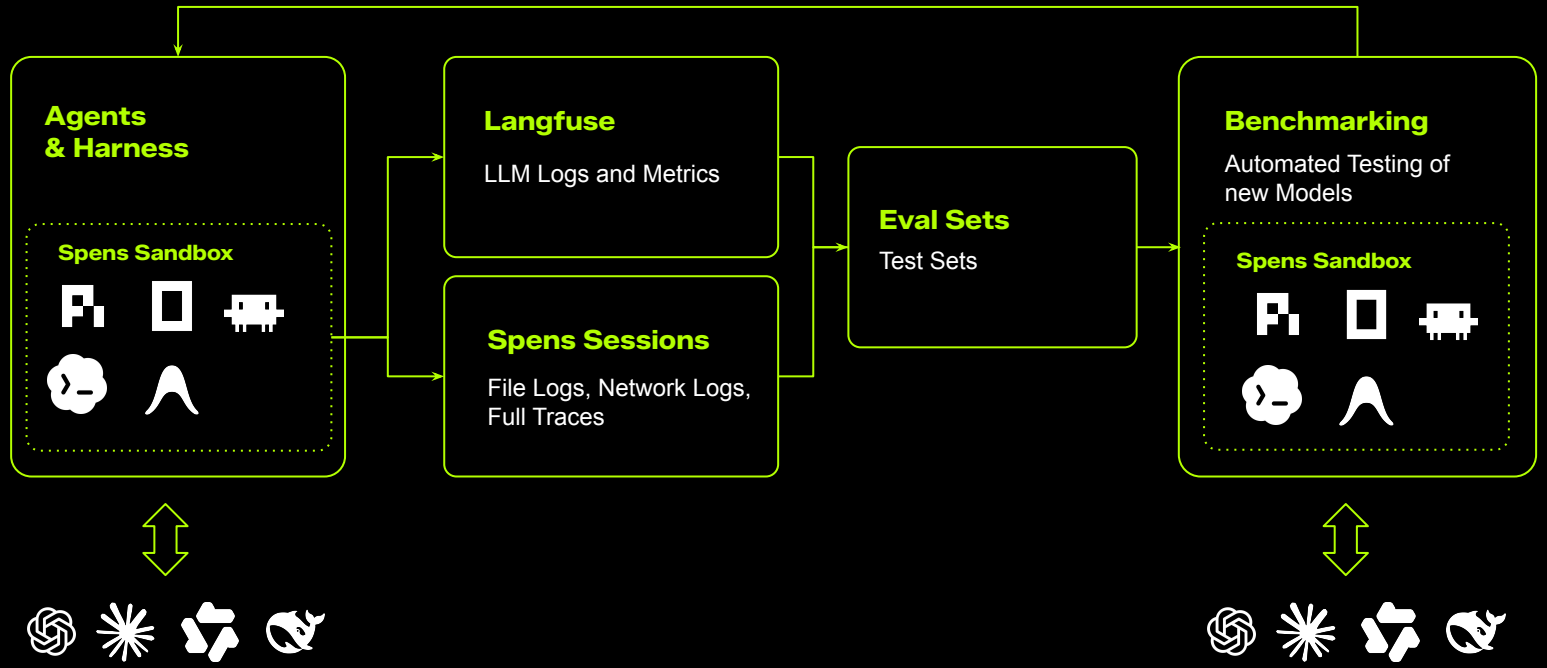
Foundational

Observability, Evals, and Benchmarking



Foundational

How we do it at ReFwd (coding agent optimization)



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Technique 1:

Just Improve the Frontier

Cache Improvements	Bulk Processing	Dynamic Tool Calls	Dynamic Compaction
Ensure caching is turned on and optimize prompt / input order to optimize cache hits	Process jobs in batch that do not require real time processing (memory compaction for instance)	Use a system 1 model / classifier or embedding model to optimize tools sent to LLM	Use a system 1 model / classifier or embedding model to optimize what context is sent to a request (pre filtering)
Up to 90% cheaper, 85% faster	50% off every token	Up to 85% fewer tool tokens, +25 pts accuracy	~60% cheaper and more accurate.

Anthropic: Message Batches API , Anthropic Engineering: Advanced tool use, Plushcap: Anthropic prompt caching, IntuitionLabs: LLM prompt caching cost savings, Introl: Prompt caching infrastructure, Simon Willison: Anthropic batch mode, Respan: Anthropic Message Batches API, Blockchain.News: Anthropic context management, Breeze in Flow: The next bottleneck is context

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Technique 2:

Dynamic routing between models

Intent Classification - Routing

Leveraging an intent classifier to dynamically route the request to models that meet those tasks

Delegation Routing

Delegating tasks to smaller more effective models, with a planner model or similar agent

Escalation Routing

Evaluating signals of failure (lack of tool call adherence, user feedback etc) to escalate to more expensive models

Case Study

Hybrid beat Opus 5.5 on pass rate and security at 21% lower cost

	Opus 5.5	Qwen	Hybrid Model (Qwen / Opus 5.5)
Completion	0.90	0.84	**0.95**
Security	0.68	0.64	**0.75**
Insec	25%	24%	**21%**
Cost	~$15.23	**~$9.15**	~$12.04

*Based on internal heavily modified baxbench across Python[FastAPI, Flask] and Javascript[Express, Koa] on pi agent running in spens sandbox over a horizon of 40 tasks with 3 samples each. Test ran October 26th.

Technique 3:

Local Models

For small specialized tasks, where you have clear outcomes or are on unreliable connections you can leverage local models moving compute to the edge and protecting privacy

Challenge	Approach	Result
Tool Call Format Adherence	Constrained decoding	100% Adherence
Call and Context Bloat	Same as foundation, compaction and selection models	13.6% → 43.1% tool selection, ~50% fewer prompt tokens (RAG-MCP)
Multi-turn drift	Agentic trajectory fine-tuning + env guardrails	8B model hits 46.7% on τ-bench

Ubicloud - Fixing Tool Call Accuracy in vLLM
RAG-MCP (arXiv 2505.03275), APIGen-MT / xLAM-2 (arXiv 2504.03601)

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Case Study

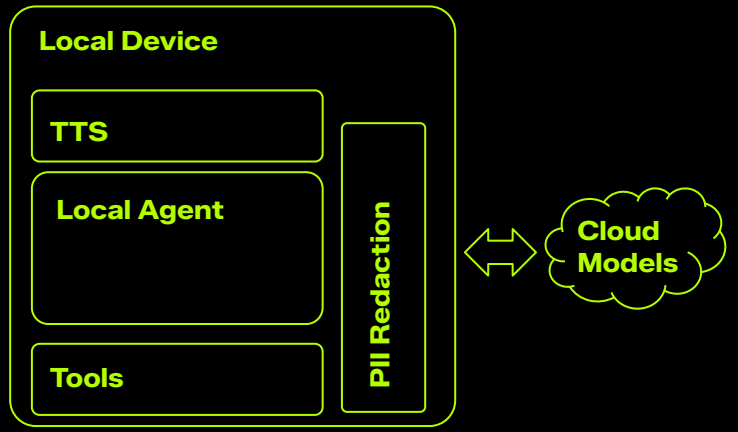
Local Home Assistant

Localized home assistant research project.

- Jetson AGX (64gig)
- Liquid AI LFM2-24B-A2B
- Whisper for STT
- ai4privacy/llama-ai4privacy-multilingual-categorical-anonymiser-openpii
- Claude Sonnet cloud

Results

- 100% Tool call format adhesion.
- PII redaction and subagent tool call, 0/427 PII leakage for search / web tasks
- 94% accuracy compared to cloud 98% accuracy but 100% task completion (more retries on local model)



Technique 4:

Fine-Tuned specialists models

Task Specific FT

Taking evals and golden sets and fine tuning models to improve performance

Student / Teacher Dynamic FT

Similar to task specific FT but leverage a larger teacher model to create the truth and train off that through Reinforcement Learning

Technique 4: Case Study

arXiv 2606.05781: Domain-Adapted SLMs with Hybrid Post-Processing · June 2026

219 examples, 2% of weights — a LoRA-tuned Llama 3.1 8B matches gpt-4o at 46–76% lower cost

83%
human-validated accuracy

100%
JSON structural validity — no malformed output vs. parsing failures from prompted LLMs

$0.013
per evaluation vs. $0.025–$0.055 for frontier APIs → **46–76% cost savings** at scale

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Recap

What can go wrong

Cheaper per turn, 10x the turns

Smaller models are cheaper per turn, but can take 10x the amount of turns in agentic workloads, monitoring is key to understand actual savings

Bad evals, Bad UX

Without good evaluations / benchmarks, swapping models can negatively impact experience

Hosting is complicated

Hosting your own model is complicated, and there are hidden costs around maintenance and monitoring.

Conclusion

Frontier by default is a design flaw

By understanding your tasks, benchmarking, and applying various strategies, you can achieve the balance of cost, speed, and accuracy

	Foundational Improvement	Dynamic Routing	Local	Fine Tuned Models
Tactics	- Cache Optimization - Batch Processing - Tool call cleanup - Context Compaction	- Intent Routing - Delegation Routing - Escalation Routing	- Constrained Decoding - Tool call cleanup - Context Compaction	- Task Specific FT - Student / Teacher FT
Good When	Always (when your task is fairly well known)	You have clean benchmarks and can evaluate	You care about data privacy or distributed compute cost	When you have narrow specific tasks and good evals and benchmarks

Evals, Observability, and Benchmarking

Conclusion

If you take one thing away it's:

Observe, Measure, Experiment -
and you too can have cost, speed, and accuracy

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Questions And Answers

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